

Module 6

Information Theory



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Waveform and Vector AWGN Channels: Equivalence

- $r(t) = s_m(t) + n(t)$, $1 \leq m \leq M$
- Vector representation: $\mathbf{r} = \mathbf{s}_m + \mathbf{n}$
- All vectors are of size N : orthogonal expansion using N basis signals.
- $s_m(t) = \sum_{j=1}^N s_{mj} \phi_j(t)$
- $n_t(t) = \sum_{j=1}^N n_j \phi_j(t)$
- Let $n_j \sim (0, N_0/2)$
- What is PDF noise vector $\mathbf{n}$?
- $f(\mathbf{n}) = \left(\frac{1}{\sqrt{\pi N_0}}\right)^N e^{-\|\mathbf{n}\|^2/N_0}$

Optimum Detector Rule: ML

- $f[\mathbf{r}|\mathbf{s}_m]$: likelihood
- If $P_m = 1/M$ (equi-probable)
- $\hat{m} = \operatorname{argmax}_{1 \leq m \leq M} f(\mathbf{r}|\mathbf{s}_m)$
- Maximum Likelihood receiver.

Optimum Detector for AWGN: ML

- $\hat{m} = \operatorname{argmax}_{1 \leq m \leq M} [P_m f[\mathbf{r} | \mathbf{s}_m]]$
- $\hat{m} = \operatorname{argmax}_{1 \leq m \leq M} [P_m f_{\mathbf{n}}(\mathbf{r} - \mathbf{s}_m)]$
- $\hat{m} = \operatorname{argmax}_{1 \leq m \leq M} [P_m (\frac{1}{\sqrt{\pi N_0}})^N e^{-\|\mathbf{r} - \mathbf{s}_m\|^2 / N_0}]$
- $\hat{m} = \operatorname{argmax}_{1 \leq m \leq M} [P_m e^{-\|\mathbf{r} - \mathbf{s}_m\|^2 / N_0}]$
- $\hat{m} = \operatorname{argmax}_{1 \leq m \leq M} [\ln P_m - \frac{\|\mathbf{r} - \mathbf{s}_m\|^2}{N_0}]$
- $\hat{m} = \operatorname{argmax}_{1 \leq m \leq M} [-\|\mathbf{r} - \mathbf{s}_m\|^2]$
- $\hat{m} = \operatorname{argmin}_{1 \leq m \leq M} [\|\mathbf{r} - \mathbf{s}_m\|^2]$
- ML receiver
- Nearest-neighbour or Minimum distance receiver or One shot receiver.
- The receiver receives $\mathbf{r}$ and looks among all $\mathbf{s}_m$ to find the one that is closest to $\mathbf{r}$ using standard Euclidean distance.

SER for M-PSK

- $D_1 = \{\theta : -\pi/M < \theta \leq \pi/M\}$
- $P_e = 1 - \int_{-\pi/M}^{\pi/M} f(\theta) d\theta$
- $M = 2, P_e = Q(\sqrt{\frac{2E_b}{N_0}})$
- $M = 4, P_e = 2Q(\sqrt{\frac{2E_b}{N_0}})[1 - \frac{1}{2}Q(\sqrt{\frac{2E_b}{N_0}})]$

SER for M-ary Orthogonal Modulation

- $s_m = (\sqrt{E}, 0, \dots, 0), \dots, (0, 0, \dots, \sqrt{E})$
- $\mathbf{r} = (\sqrt{E} + n_1, n_2, n_3, \dots, n_M)$
- Distance between two points: $\sqrt{2E}$
- Detector $R_m = \mathbf{r} \cdot \mathbf{s}_m$
- Assume s_1 is transmitted.
- $R_1 = E + \sqrt{E}n_1, R_m = \sqrt{E}n_m, 2 \leq m \leq M$
- Suppose $\mathbf{s}_1$ was transmitted.
- Correct $P_c = Prob[R_1 > R_2, R_1 > R_3, \dots, R_1 > R_m | \mathbf{s}_1]$

Objectives of Today Lecture 38

- Importance of the Module
- Information Theory: Intro
- Source Coding
- Channel Capacity
- Channel Coding

Module 6: Information Theory

How are you?

Represent the word "how" using binary representation using the following criteria.

- Encode each character separately
- Use minimum code word per character
- Code word for each character must be unique

Letter and Probability

Letter	Probability
Space	0.187
E	0.1073
T	0.0856
A	0.0668
O	0.0654
N	0.0581
R	0.0559
I	0.0519
S	0.0499
H	0.04305
D	0.03100
L	0.02775
F	0.02395
C	0.02260
M	0.02075
U	0.02010
G	0.01633
Y	0.01623
P	0.01623
W	0.01620
B	0.01179
V	0.00752
K	0.00344
X	0.00136
J	0.00108
Q	0.00099
Z	0.00063

Source Encoding: Motivation

- Using the probabilistic criteria, and encode the word "will"
- Compare with equal probability case

Measure of information I

- Quantitative measure of information?
- Lower is the probability of occurrence: higher is the information content.
- $I = \log_2(1/P) = -\log_2(P)$ bits/message

Average Information per Message: Entropy

- A source m emits random discrete messages $m_1, m_2, \dots, m_n$ with probabilities $P_1, P_2, \dots, P_n$.
- Average information per message $H(m) = \sum_i^n P_i I_i$ bits/message.
- $H(m) = - \sum_i^n P_i \log_2 P_i$ bits/message.
- Average information per message $H(m)$: entropy of the source.
- Entropy: Minimum number of bits to encode a message.

Average Code Length

- $L = \sum P_i L_i$
- P_i : Probability of i -th message
- L_i : Code length of i -th message

Prefix Condition

- Codes should be easily decodable.
- A code which satisfies prefix condition can be decoded more easily.
- Prefix condition: A longer code should not have prefix of smaller codes.
- For example, $C_1 = 011$, and $C_2 = 01111$ (does not satisfy the prefix condition)

Kraft Inequality

$$\frac{1}{2} \leq \sum_{i=1}^N 2^{-L_i} \leq 1 \quad (1)$$

L_i : Length of i -th code, N : Number of codes.

Source Coding Theorem

Theorem

The expected length L of any code for message m is greater than or equal to the entropy $H(m)$; that is, $L \geq H(m)$, and equality if $2^{-L_i} = P_i$

Efficiency and Redundancy

Efficiency $\eta = \frac{H(m)}{L}$

Redundancy $1 - \eta$

Source Coding: Example

- Message= A
- Calculate minimum number of bits to represent the message source A
- We need the probability of occurrence of A , say $P_A = 1$
- $H(m) = -\sum_i^n P_A \log_2 P_A = 0$ bits/message.

Problem 12.1-1 (TB)

- A message source generates one of four messages randomly every microsecond. The probabilities of these messages are 0.4, 0.3, 0.2, 0.1 . Each emitted message is independent of the other messages in the sequence. What is the source entropy?

Objectives of Today Lecture 39

- Variable Length Source Coding
- Channel Capacity

Source Coding: Important Results

- $L \geq H(m)$
- Prefix condition: A longer code should not have prefix of smaller codes.
-

$$\frac{1}{2} \leq \sum_{i=1}^N 2^{-L_i} \leq 1$$

L_i : Length of i -th code, N : Number of codes.

Source-Coding: Example

- How many messages?
- Number of messages = 5.
- Number of bits to represent 5 symbols?
- Number of bits = 3
- Standard binary representation (an example):
 $A = 000, B = 001, C = 101, D = 111, E = 110$
- Minimum bits $H(m) = -[-0.2 \log_2 0.2 + 0.1 \log_2 0.1 + 0.15 \log_2 0.15 + 0.3 \log_2 0.3 + 0.25 \log_2 0.25] = 2.227$ bits/message
- Code efficiency $\eta = 2.227/3 = 0.742$
- Can we improve the code efficiency?

Source Encoding Algorithms

How to encode a message such that minimum bits are required in representation.

- Shannon-Fano Encoding
- Huffman Code.
- Lampel-Ziv Source-Coding

Shannon-Fano Coding: Procedure

- Arrange symbols in descending order of probabilities.
- Partition probability space such that probabilistic weights of these spaces at every iteration are as close to equiprobable.
- Partitioning continues till the subspaces contain only one message.
- Partitions are consistently assigned binary digits 0 for upper and 1 for lower groups.
- The code is read from root node to message symbol node.
- It resembles as a tree with two branches at each partition, the last single symbol represents the leaf.

Shannon-Fano Coding: Example

- A with $P_A = 0.20$, B with $P_B = 0.10$, C with $P_C = 0.15$, D with $P_D = 0.30$, E , with $P_E = 0.25$
- Find the code words, code word length, average code-efficiency with Shannon-Fano encoding.

Solution

$$D = 00$$

$$E = 01$$

$$A = 10$$

$$C = 110$$

$$B = 111$$

Source Extension: Improving the source coding efficiency

Problem

A source emits two messages with probabilities $1/3$ and $2/3$ randomly and independently. Find the Shannon-Fano code of second-order extension of the source.

Solution

- Second order extension: new messages are four with probabilities $1/9$, $4/9$, $2/9$, $2/9$.
- Shannon-Fano Encoding:
- Codes are 0, 10, 110, 111.

Huffman Code: Procedure

- It is: fixed to variable length code. Also called compact code.
- Sort the source outputs in decreasing order of their probabilities.
- Merge the two least probable outputs into a single output whose probability is the sum of the corresponding probabilities.
- If the number of remaining outputs is 2, then go to the next step, otherwise go to the first step.
- Arbitrary assign 0 and 1 as code words for the two remaining outputs.
- If an output is the results of the merger of two outputs in a preceding step, append the current code word with a 0 and 1 to obtain the code word for the proceeding outputs and then repeat this step. If no output is preceded by another output in a preceding step, then stop.

Problem

A source emits seven messages with probabilities $1/2$, $1/4$, $1/8$, $1/16$, $1/32$, $1/64$, and $1/64$, respectively. Find the entropy of the source. Obtain the compact binary code and find the average length of the codeword. Determine the efficiency and the redundancy of the code.

Solution

- $H(m) = - \sum_{i=1}^7 P_i \log_2 P_i$
- $H(m) = 63/32$ bits/message

Solution: Compact or Huffman Code

Message	Prob.	Code	Prob.	Code	Prob.	Code	Prob.	Code	Prob.	Code	Prob.	Code
m_1	1/2	0	1/2	0	1/2	0	1/2	0	1/2	0	1/2	0
m_2	1/4	10	1/4	10	1/4	10	1/4	10	1/4	10	→ 1/2	1
m_3	1/8	110	1/8	110	1/8	110	1/8	110	→ 1/4	11		
m_4	1/16	1110	1/16	1110	1/16	1110	→ 1/8	111				
m_5	1/32	11110	1/32	11110	→ 1/16	1111						
m_6	1/64	111110	→ 1/32	11111								
m_7	1/64	111111										

Solution

- Average length
- $\bar{L} = \sum_{i=1}^7 P_i L_i = 63/32$ bits per message
- Efficiency $\eta = \frac{H(m)}{\bar{L}} = 1$
- Redundancy $1 - \eta = 0$

Huffman Code: Example

TABLE 13.1

Original Source		Reduced Sources			
Messages	Probabilities	S_1	S_2	S_3	S_4
m_1	0.30	0.30	0.30	→ 0.43	→ 0.57
m_2	0.25	0.25	→ 0.27	0.30	0.43
m_3	0.15	→ 0.18	0.25	0.27	
m_4	0.12	0.15	0.18		
m_5	0.08	0.12			
m_6	0.10				

TABLE 13.2

Original Source			Reduced Sources			
Messages	Probabilities	Code	S_1	S_2	S_3	S_4
m_1	0.30	00	0.30	00	→ 0.43	1
m_2	0.25	10	0.25	10	→ 0.27	00
m_3	0.15	010	→ 0.18	11	0.30	01
m_4	0.12	011	0.15	010	0.27	01
m_5	0.08	110	0.12	011		
m_6	0.10	111				

Problem

Consider the source symbols and their respective probabilities, as given in the following table. Find the efficiency of this code and its second order extension using Huffman encoding method.

<i>Symbol</i>	<i>Probability</i>	<i>Self Information</i>	<i>Codeword</i>
x_1	0.40	1.3219	1
x_2	0.35	1.5146	00
x_3	0.25	2.0000	01

Solution

<i>Symbol Pairs</i>	<i>Probability</i>	<i>Self Information</i>
x_1x_1	0.1600	2.6439
x_1x_2	0.1400	2.8365
x_2x_1	0.1400	2.8365
x_2x_2	0.1225	3.0291
x_1x_3	0.1000	3.3219
x_3x_1	0.1000	3.3219
x_2x_3	0.0875	3.5146
x_3x_2	0.0875	3.5146
x_3x_3	0.0625	4.0000

- $H(x) = -\sum_{k=1}^3 P_k \log_2(P_k) = 1.5589$ bits/message.
- $L = 1.60$ bits/message.
- Efficiency: $\eta = 1.5589/1.6 = 0.9743$

Solution

- Source extension:

Symbol Pairs	Probability	Self Information
x_1x_1	0.1600	2.6439
x_1x_2	0.1400	2.8365
x_2x_1	0.1400	2.8365
x_2x_2	0.1225	3.0291
x_1x_3	0.1000	3.3219
x_3x_1	0.1000	3.3219
x_2x_3	0.0875	3.5146
x_3x_2	0.0875	3.5146
x_3x_3	0.0625	4.0000

- $H(x)_{\text{new}} = -\sum_{k=1}^9 P_k \log_2(P_k) = 3.1177$ bits/ two messages.
- $H(x) = H(x)_{\text{new}}/2 = 1.5589$ bits/message.

Solution

• Huffman Encoding

Symbol pairs	Probability	Final Huffman Code.			
$x_1 x_1$	0.16 (000)	0.16 (000)	→ 0.1875 (11)	→ 0.2225 (10)	→ 0.28 (01)
$x_1 x_2$	0.14 (010)	→ 0.15 (001)	0.16 (000)	0.1875 (11)	0.2225 (10)
$x_2 x_1$	0.14 (011)	0.14 (010)	0.15 (001)	0.16 (000)	0.1875 (11)
$x_2 x_2$	0.1225 (100)	0.14 (011)	0.14 (010)	0.15 (001)	0.16 (000)
$x_1 x_3$	0.1 (101)	0.1225 (100)	0.14 (011)	0.14 (010)	0.15 (001)
$x_3 x_1$	0.1 (110)	0.1 (101)	0.14 (011)	0.14 (010)	0.15 (001)
$x_2 x_3$	0.0875 (111)	0.1 (110)	0.1225 (100)	0.14 (011)	0.14 (010)
$x_3 x_2$	0.0875 (0010)	0.0875 (111)	0.1 (101)	0.1225 (100)	0.14 (011)
$x_3 x_3$	0.0625 (0011)		0.1 (110)	0.1 (101)	
			0.31 (00)	→ 0.41 (1)	→ 0.59 (0)
			0.28 (01)	0.31 (00)	0.41 (1)
			0.2225 (10)	0.28 (01)	
			0.1875 (11)		

Solution

- Average length = $0.16 * 3 + 0.14 * 3 + 0.14 * 3 + 0.1225 * 3 + 0.1 * 3 + 0.1 * 3 + 0.0875 * 3 + 0.0875 * 4 + 0.0625 * 4 = 3.15$
- Efficiency: $\eta = 3.1177/3.15 = 0.9897$
- Source extension increases the code efficiency.

Channel Capacity: Motivation

- $P_e = k_1 Q\left(\sqrt{\frac{k_2 E_b}{N_0}}\right)$

Channel Capacity: Motivation

- To make $P_e \rightarrow 0$, $E_b \rightarrow \infty$
- $E_b = P_b/R_b$
- To increase E_b : either increase P_b or decrease R_b
- There is always a limit on P_b
- Hence to get zero-error transmission, $R_b \rightarrow 0$
- Shannon's Channel Capacity: Achieves zero-error transmission even without $R_b = 0$.
- In essence: If $R_b \leq C$, $p_e \rightarrow 0$, where C is channel capacity.

Objectives of Today Lecture 40

- Channel Capacity: Discrete Channel
- Channel Capacity: Continuous Channel

Condition for Maximum Entropy

- Intuitively?
- Maximize $H(m) = -\sum_{i=1}^n P_i \log_2 P_i$ under constraint of $\sum_{i=1}^n P_i = 1$.
- Lagrangian: $f(P_1, P_2, \dots, P_n) = -\sum_{i=1}^n P_i \log_2 P_i + \lambda \sum_{i=1}^n P_i$
- $\frac{df}{dP_j} = -P_j \left(\frac{1}{P_j}\right) \log_2 e - \log_2 P_j + \lambda = -\log_2 P_j + \lambda - \log_2 e$,
 $j = 1, 2, \dots, n$.
- Setting to zero $P_1 = P_2 = \dots = P_n = \frac{2^\lambda}{e}$
- Invoking the constraint:
- $n \frac{2^\lambda}{e} = 1$
- $P_1 = P_2 = \dots = P_n = \frac{1}{n}$.
- $H(m) = \log_2 n$ bits/symbol.

Entropy for transmission

- For M-ary channel: $H(X) = \log_2 M$ bits/symbol
- For binary channel: $H(X) = \log_2 2 = 1$ bits/symbol
- For 4-ary channel: $H(X) = \log_2 4 = 2$ bits/symbol
- For 8-ary channel: $H(X) = \log_2 8 = 3$ bits/symbol
- and so jayo

Conditional Entropy

- $H(X|Y)$: After observing Y , uncertainty in X becomes $H(X|Y)$.
- Remaining uncertainty about X after observing X .
- $H(Y|X)$: If X is known, the uncertainty in Y becomes $H(Y|X)$.

Mutual Information

- $I(X : Y) = H(X) - H(X|Y)$ bits/message or bits/symbol
- Intuitive: How known Y reduces uncertainty in X
- $I(X : Y) = H(Y) - H(Y|X)$ bits/message or bits/symbol
- Intuitive: How known X reduces uncertainty in Y

Channel Capacity C

- Channel capacity $C = \max_{P(x_i)} I(X; Y)$ bits per symbol.
- How much knowing Y reduces uncertainty in X .

Noise Less Channel

- $H(X|Y) = 0$ bits/message or bits/symbol
- There is zero uncertainty about X once Y is known.
- This does not mean $H(X) = 0$.
- X can still be random and unpredictable *before* seeing Y .
- But after seeing Y , there is no randomness left.
- Noise-less channel
- In this case: $C = \max_{P(x_i)} H(X)$ bits per symbol.

Noise Less Channel

- $H(Y|X) = 0$ bits/message or bits/symbol
- There is zero uncertainty about Y once X is known.
- This does not mean $H(Y) = 0$.
- Y can still be random and unpredictable *without knowing* X .
- But after knowing X , there is no randomness left.
- Noise-less channel
- In this case: $C = \max_{P(x_i)} H(Y)$ bits per symbol.

Noisiest Channel

- $H(X|Y) = H(X)$ bits/message or bits/symbol
- Y is fully independent of X
- $H(Y|X) = H(Y)$ bits/message or bits/symbol
- Y is fully independent of X
- $C = 0$

Discrete Memory Less Channel: Binary

Find the channel capacity of the channel as shown in Figure 1:

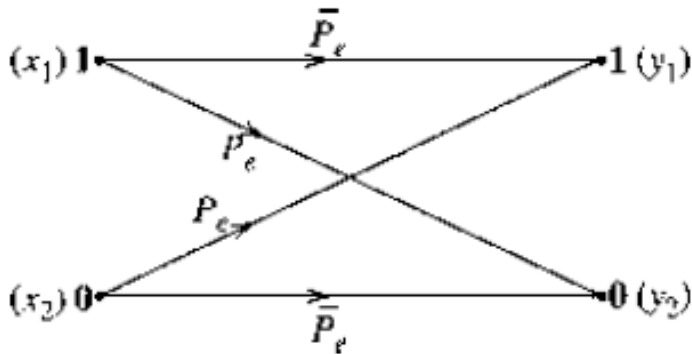


Discrete Memory Less Channel: M-ary

Find the channel capacity of an n - input and n -output channel as shown in the following figure.



Binary Symmetric Channel (BSC)



Channel Transition Matrix

$$\begin{array}{c} \text{Inputs} \end{array} \begin{array}{c} \begin{array}{c} x_1 \\ x_2 \\ \vdots \\ x_r \end{array} \end{array} \begin{array}{c} \text{Outputs} \\ \begin{array}{cccc} y_1 & y_2 & \cdots & y_s \end{array} \\ \left(\begin{array}{cccc} P(y_1|x_1) & P(y_2|x_1) & \cdots & P(y_s|x_1) \\ P(y_1|x_2) & P(y_2|x_2) & \cdots & P(y_s|x_2) \\ \vdots & \vdots & \ddots & \vdots \\ P(y_1|x_r) & P(y_2|x_r) & \cdots & P(y_s|x_r) \end{array} \right) \end{array}$$

Bayes' Rules

$$\begin{aligned} P(x_i|y_j) &= \frac{P(y_j|x_i)P(x_i)}{P(y_j)} \\ &= \frac{P(y_j|x_i)P(x_i)}{\sum_i P(x_i, y_j)} \\ &= \frac{P(y_j|x_i)P(x_i)}{\sum_i P(x_i)P(y_j|x_i)} \end{aligned}$$

Entropy:

- Entropy: $H(X) = \sum_i P(x_i) \log_2 \frac{1}{P(x_i)}$ bits per message
- Entropy: $H(Y) = \sum_j P(y_j) \log_2 \frac{1}{P(y_j)}$ bits per message
- Joint Entropy $H(X, Y) = \sum_i \sum_j P(x_i, y_j) \log_2 \frac{1}{P(x_i, y_j)}$
- Conditional entropy: $H(X|Y) = \sum_i \sum_j P(x_i, y_j) \log_2 \frac{1}{P(x_i|y_j)}$ bits per message
- Conditional entropy: $H(Y|X) = \sum_j \sum_i P(y_j, x_i) \log_2 \frac{1}{P(y_j|x_i)}$ bits per message

Mutual Information $I(X; Y)$

- Mutual information:

$$I(X : Y) = H(X) - H(X|Y) = H(Y) - H(Y|X)$$

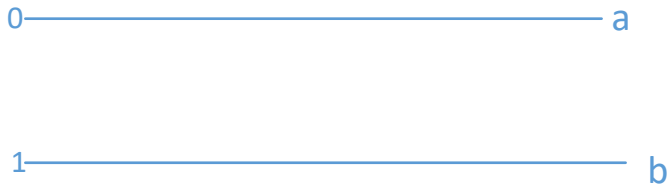
bits/message or bits/symbol

- $I(X : Y) = \sum_i \sum_j P(x_i, y_j) \log_2 \frac{P(x_i, y_j)}{P(x_i)P(y_j)}$
- $I(X : Y) = \sum_i \sum_j P(x_i)P(y_j|x_i) \log_2 \frac{P(y_j|x_i)}{\sum_i P(x_i)P(y_j|x_i)}$
- Channel capacity $C = \max_{P(x_i)} I(X; Y)$ bits per symbol.

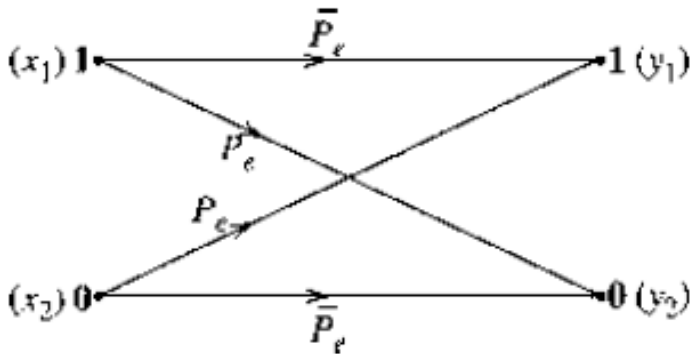
Some Identities

- $H(X, Y) = H(Y) + H(X|Y) = H(X) + H(Y|X)$
- $H(X, Y) \leq H(X) + H(Y)$
- $H(Y|X) \leq H(Y)$ and $H(X|Y) \leq H(X)$

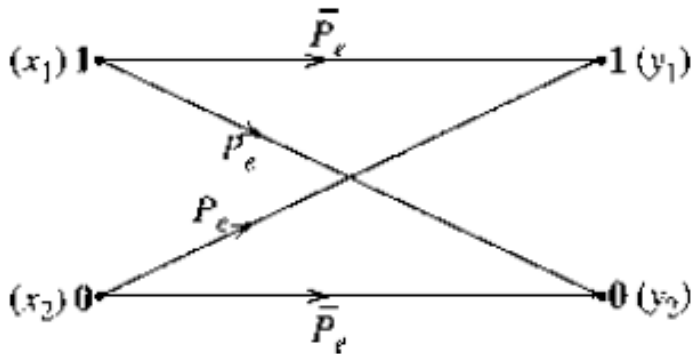
Transition Matrix



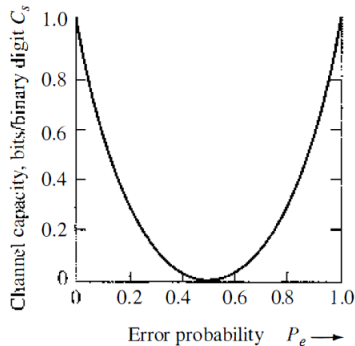
Transition Matrix for BSC



Channel Capacity: BSC



Solution: BSC Example 13.3 in TB



$$C_s = 1 - [P_e \log_2 \frac{1}{P_e} + (1 - P_e) \log_2 (\frac{1}{1 - P_e})]$$

Problem 1 TB 12.4-1

A binary channel matrix is given by

$$\begin{bmatrix} 2/3 & 1/3 \\ 1/10 & 9/10 \end{bmatrix}$$

Also $P_x(x_1) = 1/3$, and $P_x(x_2) = 2/3$. Determine $H(x)$, $H(x|y)$, $H(y)$, $H(y|x)$ and $I(x; y)$.

Solution : Understanding the Channel Matrix

- $P_{y/x}(y_1/x_1) = 2/3, P_{y/x}(y_2/x_1) = 1/3, P_{y/x}(y_1/x_2) = 1/10,$
 $P_{y/x}(y_2/x_2) = 9/10.$

Solution 1: Probability of x and y

- $P(x_1) = 1/3$, and $P(x_2) = 2/3$
- Total probability formula:
$$P(y_1) = P(y_1/x_1)P(x_1) + P(y_1/x_2)P(x_2) = 13/45$$
- $P(y_2) = 1 - 13/45 = 32/45$

Solution 1: Entropy

- $H(x) = P(x_1) \log_2 1/P(x_1) + P(x_2) \log_2 1/P(x_2) = 0.918$
bits/message
- $H(y) = P(y_1) \log_2 1/P(y_1) + P(y_2) \log_2 1/P(y_2) = 0.8673$
bits/message

Solution 1: Conditional Probabilities

Bayes' rule

- $P(x_1|y_1) = \frac{P(y_1|x_1)P(x_1)}{P(y_1)} = 10/13$

- $P(x_1|y_2) = \frac{P(y_2|x_1)P(x_1)}{P(y_2)} = 5/32$

- $P(x_2|y_1) = \frac{P(y_1|x_2)P(x_2)}{P(y_1)} = 3/13$

- $P(x_2|y_2) = \frac{P(y_2|x_2)P(x_2)}{P(y_2)} = 54/64$

Solution 1: Conditional Entropy

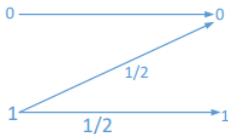
- $H(x|y_1) = P(x_1|y_1) \log_2(1/P(x_1|y_1)) + P(x_2|y_1) \log_2(1/P(x_2|y_1)) = 0.779$
- $H(x|y_2) = P(x_1|y_2) \log_2(1/P(x_1|y_2)) + P(x_2|y_2) \log_2(1/P(x_2|y_2)) = 0.624$
- $H(x|y) = P(y_1)H(x|y_1) + P(y_2)H(x|y_2) = 0.6687$

Solution 1: Mutual information

- $I(x; y) = H(x) - H(x|y) = 0.918 - 0.6687 = 0.2489.$
- $H(y|x) = H(y) - I(x; y) = 0.618$ bits/message

Problem

Find the channel capacity of Z channel:



Solution

Let $P(X = 1) = p$, $P(X = 0) = 1 - p$. $H(Y) = -(1 - p + p/2) \log_2(1 - p + p/2) - (p/2) \log_2(p/2) = -(1 - p/2) \log_2(1 - p/2) - (p/2) \log_2 p/2$.

$H(Y/X) = (-p/2) \log_2 1/2 - (p/2) \log_2(1/2)$.

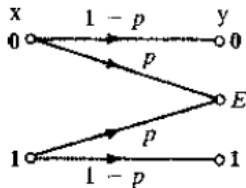
$I(X; Y) = H(Y) - H(Y/X)$.

To find p , we differentiate with $I(X; Y)$ and equate to zero. This gives $p = 2/5$. Using this, capacity is 0.322 bits/ symbol.

Problem

In data communication using error detection code, as soon as an error is detected, an automatic request for retransmission (ARQ) enables retransmission of the data in error. In such a channel, the data in error is erased. Hence, there is an erase probability p , but the probability of error is zero. Such a channel, known as a binary erasure channel (BEC), can be modeled as shown in Figure.

Determine $H(x)$, $H(x|y)$, and $I(x; y)$ assuming the two transmitted messages equiprobable.



Solution: Channel Matrix

	y_1	y_2	y_3
x_1	q	0	p
x_2	0	q	p

$$x_1 = 0, x_2 = 1, y_1 = 0, y_2 = 1, y_3 = E, q = 1 - p.$$

Solution: $H(x)$

- Symbols are equiprobable $P(x_0) = P(x_1) = 1/2$
- $H(x) = 1$ bits/symbol.

Solution: $H(x|y)$

- $H(X|Y) = \sum_i \sum_j P(x_i, y_j) \log_2 \frac{1}{P(x_i|y_j)}$ bits per symbol
- $i = 2, j = 3$
- It is for loop
- $H(x) = P(x_1, y_1) \log_2 \frac{1}{P(x_1|y_1)} + P(x_1, y_2) \log_2 \frac{1}{P(x_1|y_2)} +$
 $P(x_1, y_3) \log_2 \frac{1}{P(x_1|y_3)} + P(x_2, y_1) \log_2 \frac{1}{P(x_2|y_1)} +$
 $P(x_2, y_2) \log_2 \frac{1}{P(x_2|y_2)} + P(x_2, y_3) \log_2 \frac{1}{P(x_2|y_3)}$
- Calculate $P(x_i|y_j)$ and $P(x_i, y_j)$ for each combinations of i and j
- Use $P(x_i|y_j) = \frac{P(y_i|x_j)P(x_i)}{P(y_i)}$
- Use $P(x_i, y_j) = P(y_i|x_j)P(x_i)$
- $P(y_i|x_j)$: given from channel matrix
- $P(x_i)$: given from input probabilities (1/2 in this case)
- $P(y_j)$? from total probability theorem

Solution: $P(y_j)$

- Total probability
- $P(y_1) = P(y_1|x_1)P(x_1) + P(y_1|x_2)P(x_2) = q \times 1/2 + 0 \times 1/2 = q/2$
- $P(y_2) = P(y_2|x_1)P(x_1) + P(y_2|x_2)P(x_2) + 0 \times 1/2 + q \times 1/2 = q/2$
- $P(y_3) = 1 - q/2 - q/2 = 1 - q = p$
- $P(y_1|x_1)$ and $P(y_1|x_2)$, $P(y_2|x_1)$ and $P(y_2|x_2)$ are taken from channel transition matrix.

Solution: $P(x_i|y_j)$

- $P(x_i|y_j) = \frac{P(y_i|x_j)P(x_i)}{P(y_i)}$
- $P(x_1|y_1) = \frac{P(y_1|x_1)P(x_1)}{P(y_1)} = (q \times 1/2)/(q/2) = 1$
- Similarly
- $P(x_2|y_1) = 0$
- $P(x_1|y_2) = 0$
- $P(x_2|y_2) = 1$
- $P(x_1|y_3) = 1/2$
- $P(x_2|y_3) = 1/2$

Solution: Finally: $P(x_i, y_j)$

- $P(x_i, y_j) = P(y_i|x_j)P(x_i)$
- $P(x_1, y_1) = P(y_1|x_1)P(x_1) = q \times 1/2 = q/2$
- Similarly
- $P(x_1, y_2) = 0$
- $P(x_1, y_3) = p/2$
- $P(x_2, y_1) = 0$
- $P(x_2, y_2) = q/2$
- $P(x_2, y_3) = p/2$

Solution: $H(x|y)$

- $H(X|Y) = \sum_i \sum_j P(x_i, y_j) \log_2 \frac{1}{P(x_i|y_j)}$ bits per symbol
- $i = 2, j = 3$
- It is for loop
- $H(x) = P(x_1, y_1) \log_2 \frac{1}{P(x_1|y_1)} + P(x_1, y_2) \log_2 \frac{1}{P(x_1|y_2)} +$
 $P(x_1, y_3) \log_2 \frac{1}{P(x_1|y_3)} + P(x_2, y_1) \log_2 \frac{1}{P(x_2|y_1)} +$
 $P(x_2, y_2) \log_2 \frac{1}{P(x_2|y_2)} + P(x_2, y_3) \log_2 \frac{1}{P(x_2|y_3)}$
- $H(x|y) = p$

Solution: $I(x; y)$

- $I(x; y) = H(x) - H(x|y) = 1 - p$ bits per symbol.
- Additional questions: How to get $I(y; x)$, and how to get $H(y|x)$
- $I(y; x) = I(x; y)$
- $I(y; x) = H(y) - H(y|x)$
- $H(y|x) = H(y) - I(y; x)$
- $H(y)$ can be computed easily from probabilities of y :
 $P(y_1) = q/2$, $P(y_2) = q/2$, and $P(y_3) = p$.

Shannon's Channel Capacity for AWGN Channel

$$y = x + n$$

Differential Entropy

- Differential Entropy: $H(X) = \int_{-\infty}^{\infty} f_X(x) \log_2 \frac{1}{f_X(x)} dx$ bits per message
- Differential Conditional entropy:
 $H(X|Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y) \log_2 \frac{1}{f_{X|Y}(x|y)} dx dy$ bits per message

Mutual Information $I(X; Y)$: Continuous

- Mutual information:

$$I(X : Y) = H(X) - H(X|Y) = H(Y) - H(Y|X)$$

- $I(X : Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y) \log_2 \frac{f_{X,Y}(x,y)}{f_X(x)f_Y(y)} dx dy$ bits per message

Entropy: Uniform Distribution

- $f(x) = \frac{1}{2M}$ in the range $-M < x < M$ otherwise 0
- $H(X) = \int_{-\infty}^{\infty} f_X(x) \log_2 \frac{1}{f_X(x)} dx$ bits per message
- $H(X) = \int_{-\infty}^{\infty} f_X(x) \log_2 \frac{1}{f_X(x)} dx$
- $H(X) = \int_{-M}^M \frac{1}{2M} \log_2 \frac{2M}{1} dx = \log_2 2M$

Entropy: Gaussian Distribution

- $f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x^2/2\sigma^2}$
- $H(X) = \int_{-\infty}^{\infty} f_X(x) \log_2 \frac{1}{f_X(x)} dx$ bits per message
- Simplify $\log_2(\cdot)$ and use the property of PDF and variance
- $H(x) = \int_{-\infty}^{\infty} f_X(x) \log_2 \frac{1}{f_X(x)} dx = \frac{1}{2} \log_2(2\pi e\sigma^2)$

SOLUTION

$$H(x) = \int_{-\infty}^{\infty} p(x) \log_2 \frac{1}{p(x)} dx$$

Note that

$$\begin{aligned} \log \frac{1}{p(x)} &= \log \left(\sqrt{2\pi\sigma^2} e^{x^2/2\sigma^2} \right) \\ &= \frac{1}{2} \log (2\pi\sigma^2) + \frac{x^2}{2\sigma^2} \log e \end{aligned}$$

Hence,

$$\begin{aligned} H(x) &= \int_{-\infty}^{\infty} p(x) \left[\frac{1}{2} \log (2\pi\sigma^2) + \frac{x^2}{2\sigma^2} \log e \right] dx & (13.42a) \\ &= \frac{1}{2} \log (2\pi\sigma^2) \int_{-\infty}^{\infty} p(x) dx + \frac{\log e}{2\sigma^2} \int_{-\infty}^{\infty} x^2 p(x) dx \end{aligned}$$

Condition for Maximum Entropy

- Find $f_X(x)$ that maximizes $H(x) = \int_{-\infty}^{\infty} f_X(x) \log_e \frac{1}{f_X(x)} dx$
- with a constraint that
- Constraint 1: $\int_{-\infty}^{\infty} f_X(x) dx = 1$
- Any other constraint?
- Input signal should have a limited power
- Constraint 2: $\int_{-\infty}^{\infty} x^2 f_X(x) dx = \sigma^2$
- **Main result:** for a given mean square value, the entropy is maximum for a Gaussian distribution and the maximum entropy is $\frac{1}{2} \log_2(2\pi e \sigma^2)$.

Channel Capacity of Band-Limited AWGN

- $y = x + n$
- $H(y|x)$?
- $H(y|x) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y) \log_2 \frac{1}{f_{Y|X}(y|x)} dx dy$ bits per message

Channel Capacity of Band-Limited AWGN

- $H(y|x) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y) \log_2 \frac{1}{f_{Y|X}(y|x)} dx dy$
- Bayes' Rule:
$$H(y|x) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_X(x) f_{Y|X}(y|x) \log_2 \frac{1}{f_{Y|X}(y|x)} dx dy$$
- $H(y|x) = \int_{-\infty}^{\infty} f_X(x) dx \int_{-\infty}^{\infty} f_{Y|X}(y|x) \log_2 \frac{1}{f_{Y|X}(y|x)} dy$
- $H(y|x) = \int_{-\infty}^{\infty} f_{Y|X}(y|x) \log_2 \frac{1}{f_{Y|X}(y|x)} dy$
- $f_{Y|X}(y|x) = f_n(y - x)$
- $H(y|x) = \int_{-\infty}^{\infty} f_n(y - x) \log_2 \frac{1}{f_n(y-x)} dy$
- Let $z = y - x$
- $H(y|x) = \int_{-\infty}^{\infty} f_n(z) \log_2 \frac{1}{f_n(z)} dz$
- $H(y|x) = H(n)$

Channel Capacity of Band-Limited AWGN

- $I(x; y) = H(y) - H(y|x) = H(y) - H(n)$
- Capacity: For a given $H(n)$, $I(x; y)$ is maximum when $H(y)$ is maximum.
- $y = x + n$
- $\bar{x}^2 = S$, $\bar{n}^2 = N$
- For a given mean square value $\bar{y}^2 = S + N$, $H(y)$ is maximum when y is Gaussian.
- Maximum $H(y)$: $\frac{1}{2} \log_2[2\pi e(S + N)]$
- If n is Gaussian, y will be Gaussian only if x is Gaussian.
- $f_X(x) = \frac{1}{\sqrt{2\pi S}} e^{-x^2/2S}$.

Shannon Channel Capacity of Band-Limited AWGN

- $I(x; y) = H(y) - H(y|x) = H(y) - H(n)$
- Maximum $H(y)$: $\frac{1}{2} \log_2[2\pi e(S + N)]$
- For Gaussian noise: $H(n) = \frac{1}{2} \log_2[2\pi eN]$, $N = N_0B$.
- For signal: best is Gaussian
- For noise: worst is Gaussian
- $C = \frac{1}{2} \log_2[2\pi e(S + N)] - \frac{1}{2} \log_2[2\pi eN]$
- $C = \frac{1}{2} \log_2[1 + \frac{S}{N}]$ bits per symbol
- For a BW of B: maximum 2B symbols per second (Nyquist ISI criteria)
- $C = 2B \frac{1}{2} \log_2[1 + \frac{S}{N}] = B \log_2(1 + S/N) = B \log_2(1 + SNR)$

Shannon Channel Capacity of Band-Limited AWGN

- $C = B \log_2(1 + SNR)$
- Assumptions to get the above?
- Channel: Band-limited
- Noise: Additive
- Noise: Gaussian distributed
- Input signals: White Gaussian

Shannon Channel Capacity: Asymptotic

- Capacity when transmitted signal power $\rightarrow \infty$?
- $C \rightarrow \infty$
- Capacity when channel BW $\rightarrow \infty$?
- $C = B \log_2(1 + \frac{S}{N})$
- $C = B \log_2(1 + \frac{S}{N_0 B})$, where N_0 : PSD, N : Power.
- $\lim_{B \rightarrow \infty} C = \lim_{B \rightarrow \infty} B \log_2(1 + \frac{S}{N_0 B})$
- $\lim_{B \rightarrow \infty} C \lim_{B \rightarrow \infty} \frac{S}{N_0} [\frac{N_0 B}{S} \log_2(1 + \frac{S}{N_0 B})]$
- $\lim_{B \rightarrow \infty} C = \frac{S}{N_0} \log_2 e = 1.44 \frac{S}{N_0}$

Problem

Find the channel capacity for an AWGN channel with $B = 10$ MHz, noise PSD -120 dBW/Hz, and signal power 20 dBm. What will be channel capacity if $B \rightarrow \infty$.

Solution

- AWGN channel capacity $C = B \log_2(1 + SNR)$: everything in linear and SI units.
- $B = 10^7$ Hz
- Noise PSD: $-120\text{dBW/Hz} = 10^{-12}$ W/Hz
- Noise power $10^{-12} \times 10^7 = 10^{-5}$ Watt
- Signal Power = 20 dBm = 10^2 mW = 0.1 Watt.
- SNR = Signal power/Noise Power = $0.1/10^{-5} = 10^6 = 60\text{dB}$
- $C = B \log_2(1 + SNR) = 10^7 \log_2(1 + 10^6) = 1.99 \times 10^8$ bps = 200 Mbps.
- $C_\infty = 1.44 \frac{S}{N_0} = 10^{11}$ bps = 100 Gbps